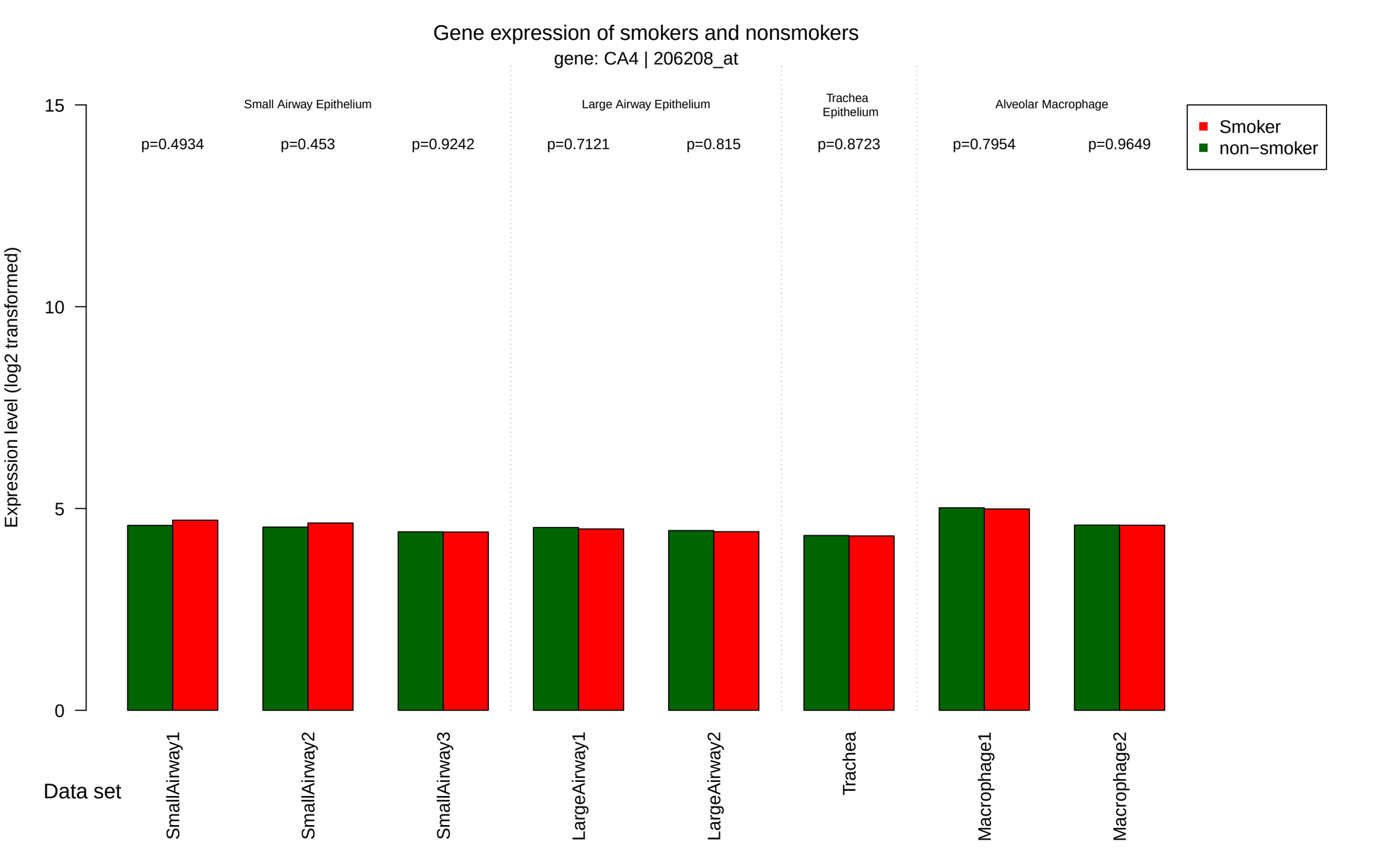
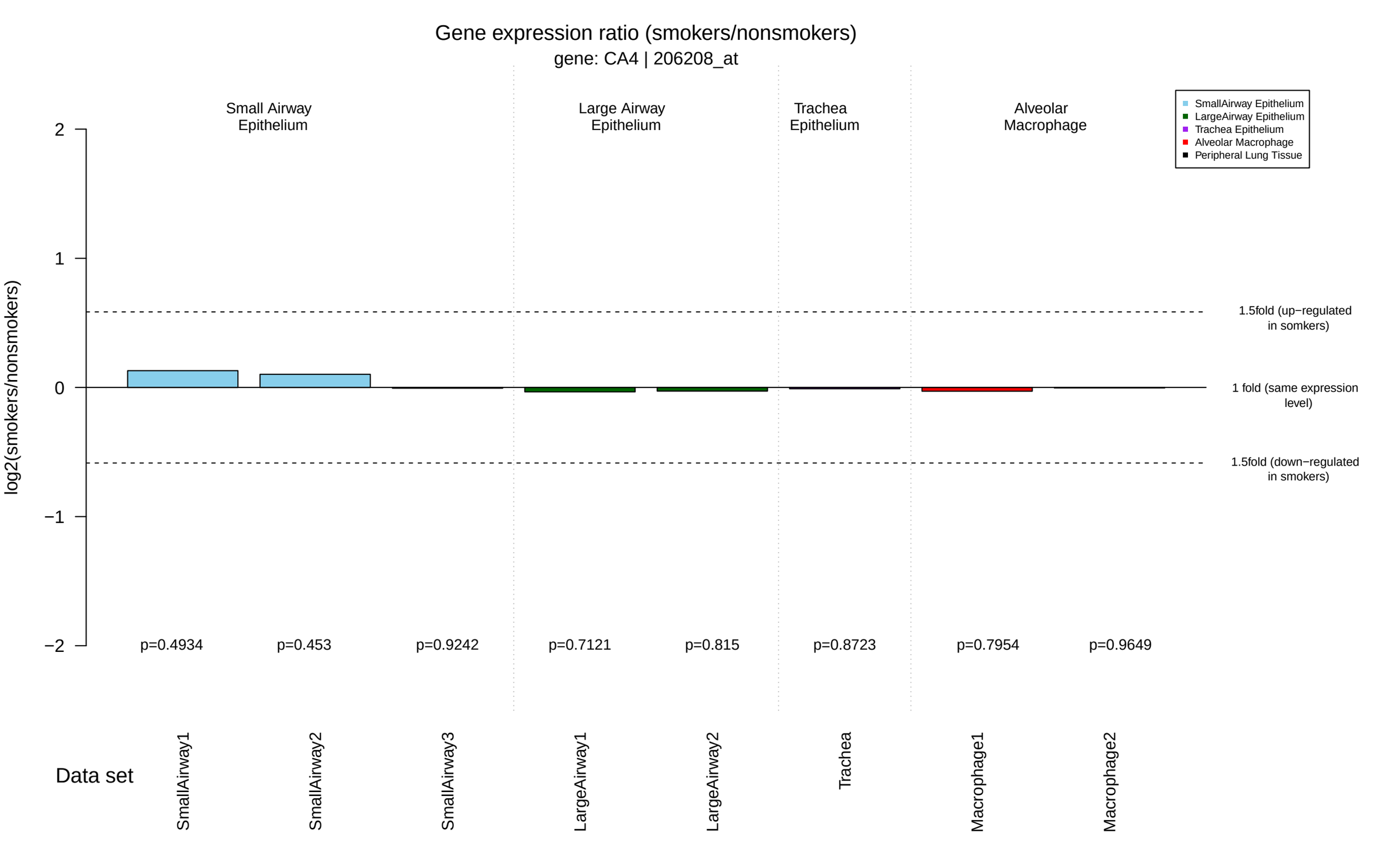
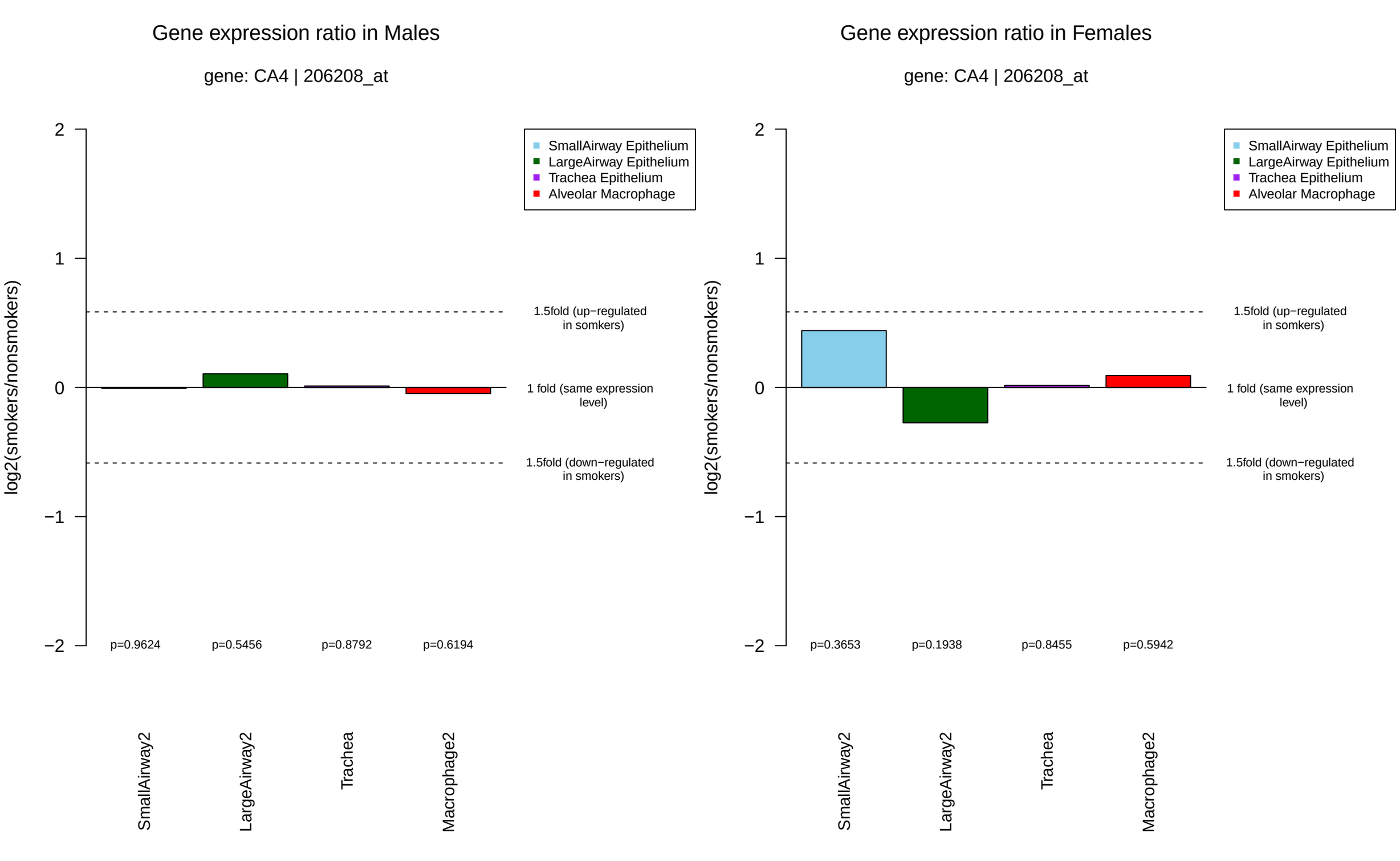
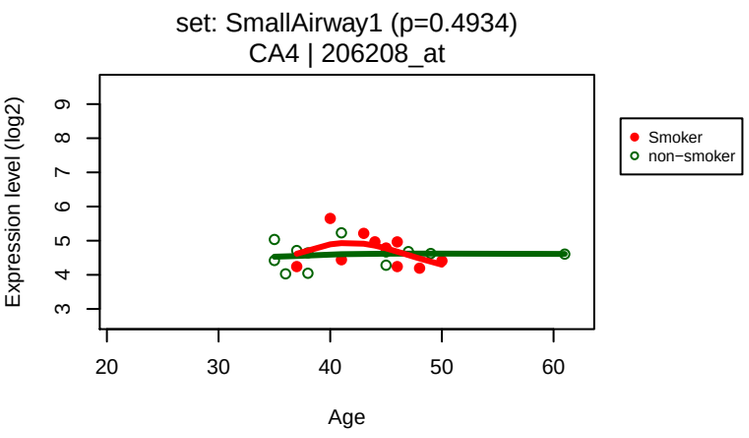
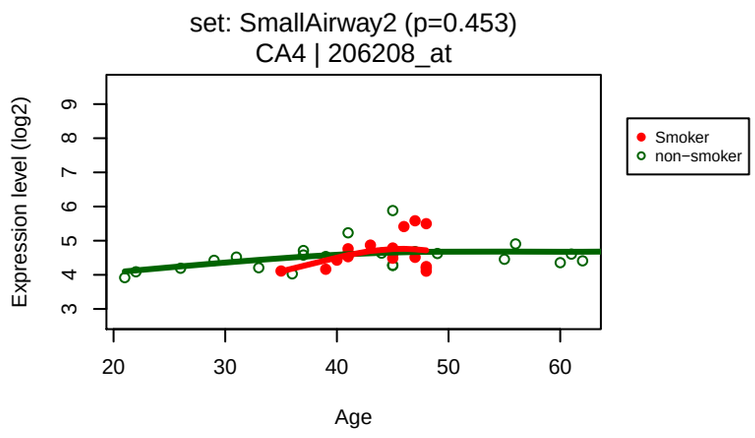
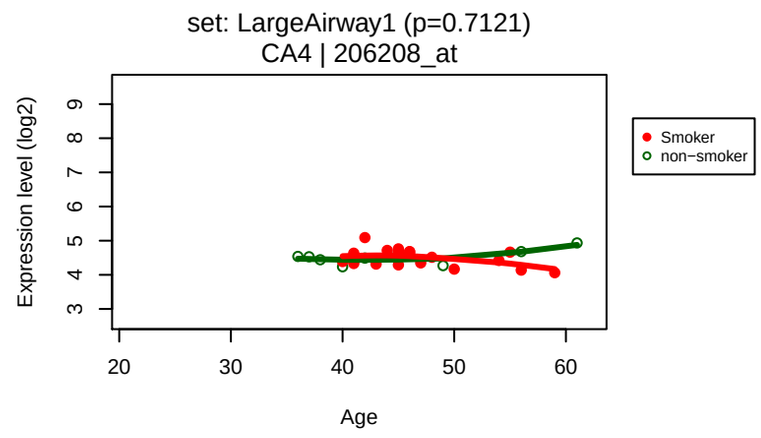

Gene expression of smokers and nonsmokers
gene: CA4 | 206208_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4934
p=0.453
p=0.9242
p=0.7121
p=0.815
p=0.8723
p=0.7954
p=0.9649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

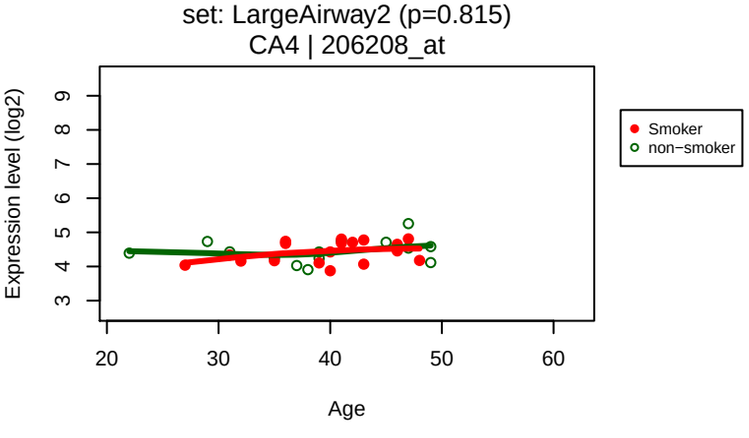

Gene expression ratio (smokers/nonsmokers)
gene: CA4 | 206208_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4934
p=0.453
p=0.9242
p=0.7121
p=0.815
p=0.8723
p=0.7954
p=0.9649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

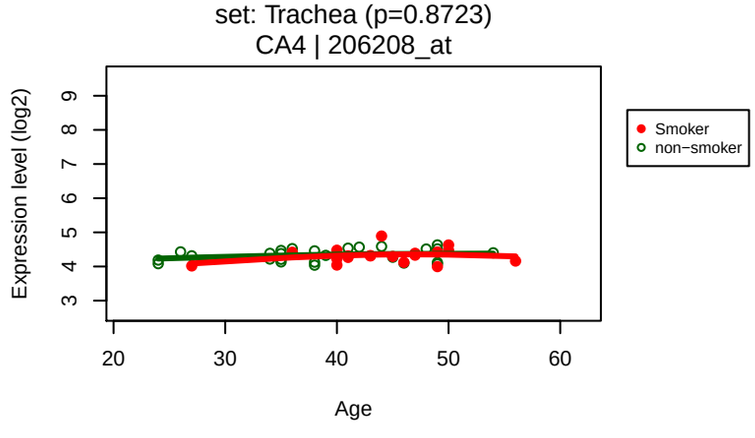

Gene expression ratio in Males
gene: CA4 | 206208_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9624
p=0.5456
p=0.8792
p=0.6194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CA4 | 206208_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3653
p=0.1938
p=0.8455
p=0.5942
SmallAirway2
LargeAirway2
Trachea
Macrophage2

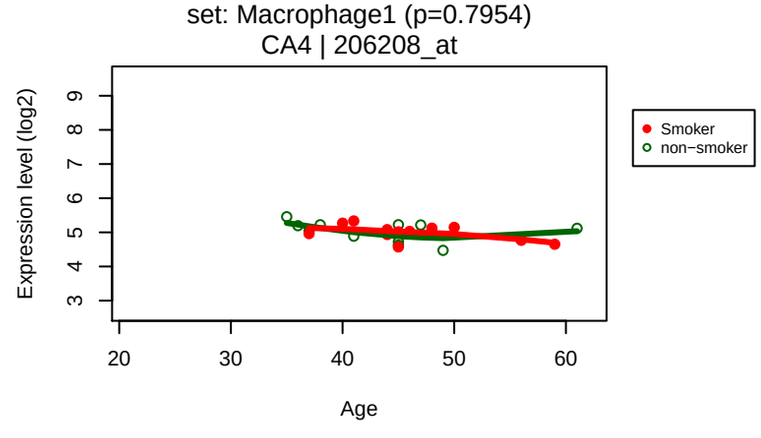

set: SmallAirway1 (p=0.4934)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker

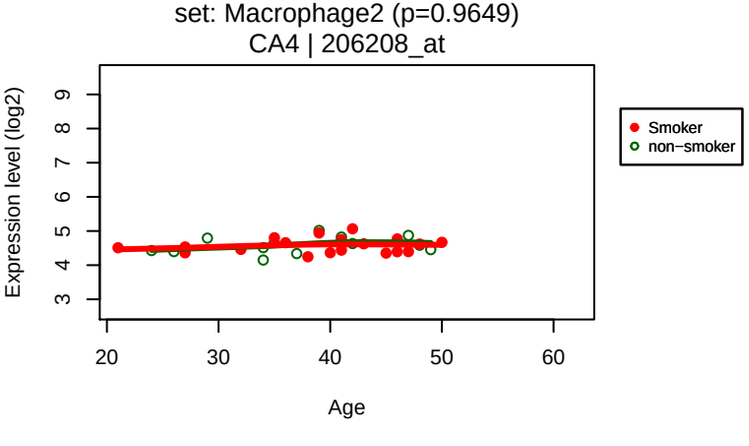

set: SmallAirway2 (p=0.453)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker

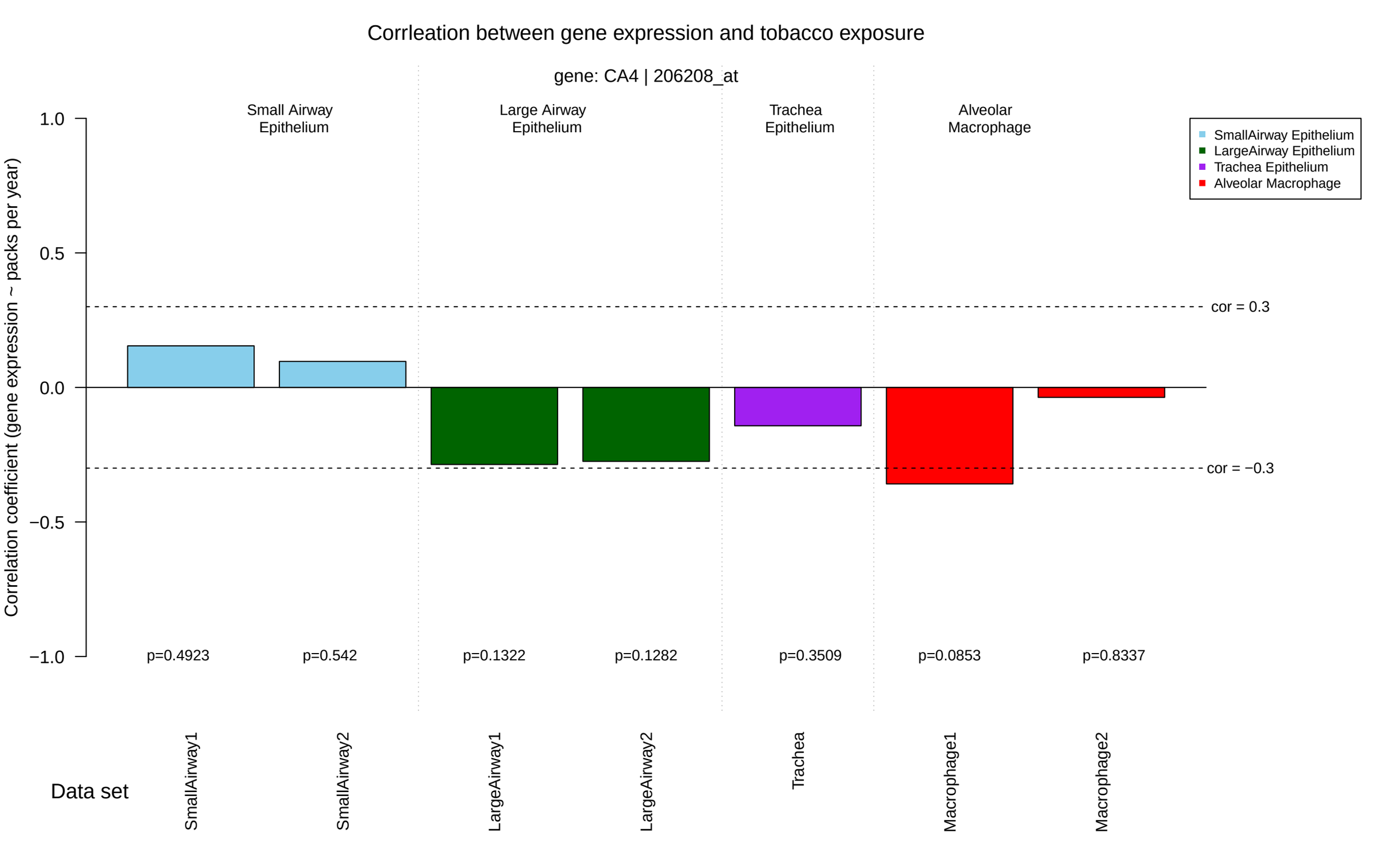

set: LargeAirway1 (p=0.7121)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.815)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8723)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7954)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9649)
CA4 | 206208_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CA4 | 206208_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4923
p=0.542
p=0.1322
p=0.1282
p=0.3509
p=0.0853
p=0.8337
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2